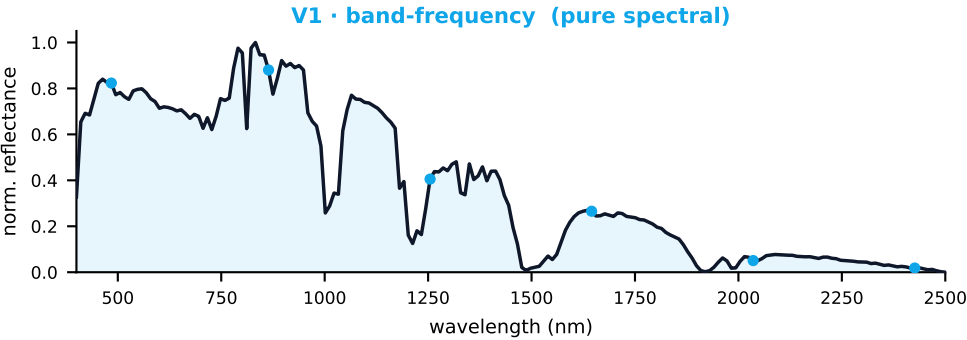


One spectrum, four token bags: the wordification step of LDA-on-HSI

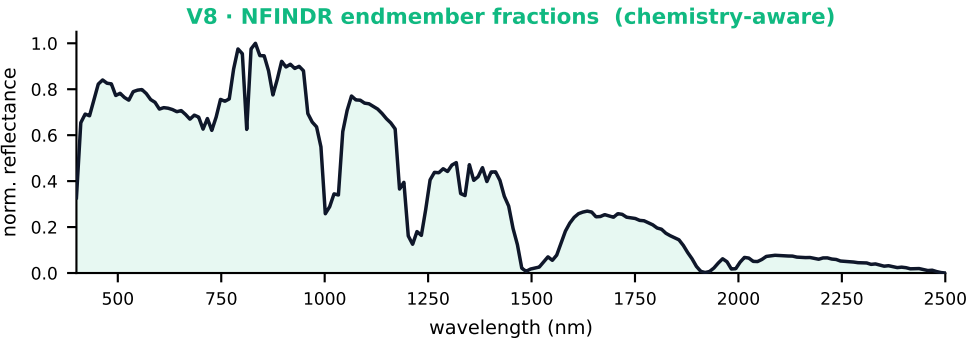
The identical pixel spectrum (left of each row) is turned into a bag-of-tokens by four recipes, one per design family. Source: stored document of a sampled Soybean-notill pixel of Indian Pines (row 142, column 30; 200 bands, 400-2500 nm).



token = band id · count = quantised intensity (0..7) · vocab = B = 200 bands



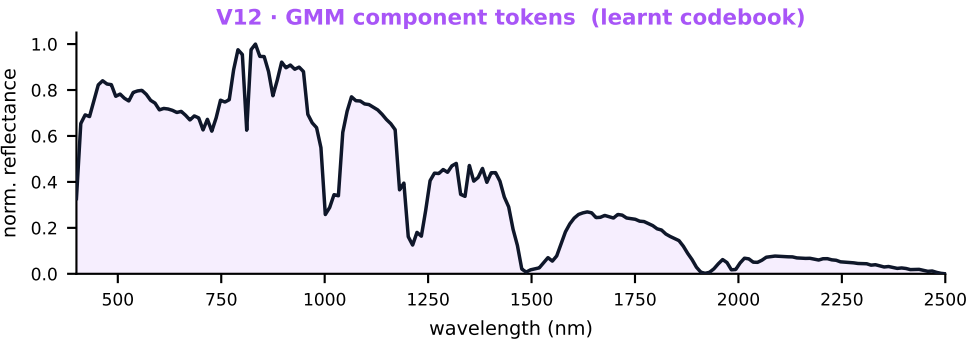
Each band token is repeated as often as its intensity bin; this document has 499 tokens. The Procemin 2022 band-frequency recipe, the canonical baseline.



NFINDR gives K=12 endmembers; NNLS unmixing gives abundance fractions. token = endmember id, count = abundance / max abundance in Q bins
abundance fractions (sum = 1)



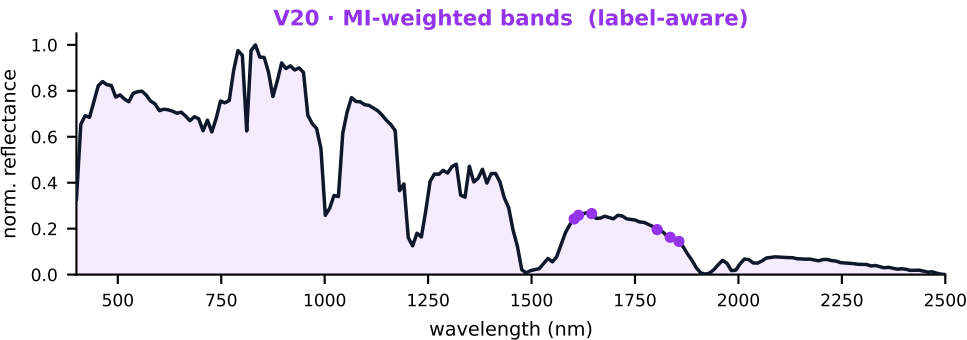
3 of 12 endmembers emit tokens in this pixel (16 tokens): the bag is short and chemistry-anchored.



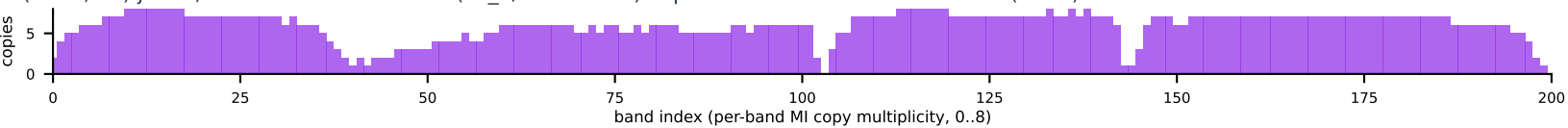
one 1-D GMM (Q=8) fitted to the raw band intensities of all sampled pixels; each band value takes its most probable component g. token = (band, g)
bands per GMM component (B=200 total)



The components are shared by all pixels and follow the intensity distribution; V3 instead splits each spectrum's own min-max range into Q equal bins.



V3 (band,bin) joints, but band b emits $\text{round}(\text{MI}_b / \text{maxMI} \times 8)$ copies. 1 of 200 bands fall silent ($\text{MI} \approx 0$).



Copies run from 0 to 8 (mean 5.9): low-MI bands are down-weighted and 1 band is silent. The only label-aware recipe in the sweep.

All four bags come from the same pixel. V1 and V20 keep the band grid; V8 collapses to the endmember simplex; V12 replaces per-spectrum equal bins with a mixture fitted to all sampled pixels; V20 re-weights by label mutual information. Tokens are the stored Q = 8 documents of this pixel in the V-sweep corpus.